SPLITTING OFF THE QUASI-FINITE PART AFTER AN ETALE NEIGHBOURHOOD, FOR ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7494-7541, immediately following the statement at 7471-7492.

Editorial note. Ten cross-chapter pointers (Decent Spaces, Groupoids, More on Morphisms, Morphisms, Spaces, Morphisms of Spaces) are recorded in assembly.external_references.

Editorial note. The proof's final steps (affineness of V from a finite flat equivalence relation, then finiteness of V -> U) are displayed inside the span.

Allowed standard prerequisites (editorial).

- The technical schemes-level splitting statement. More on Morphisms, lemma more-morphisms-lemma-etale-splits-off-quasi-finite-part-technical, cited in the proof. Supplies the scheme-level construction that is transported to algebraic spaces here.
- Presentation of an algebraic space as a quotient by an etale equivalence relation. Spaces, lemma spaces-lemma-space-presentation, cited at proof lines 7528-7530. Realises the intermediate algebraic space V as a quotient of Z_1 by R = Z_1 x_V Z_1.
- Affineness of a quotient by a finite flat equivalence relation. Groupoids, proposition groupoids-proposition-finite-flat-equivalence, cited at proof lines 7534-7535. Concludes that V is an affine scheme from finiteness of the projections of R.
- A proper morphism with finite image is finite. Morphisms, lemmas morphisms-lemma-image-proper-is-proper and morphisms-lemma-finite-proper, cited at proof lines 7536-7539. Turns properness of V -> U into finiteness, which is the conclusion.
- Finiteness of fibres and properness results for decent spaces. Decent Spaces lemmas (decent-finite-fibre, qf-and-qc-finite-fibre, section-reasonable-decent) and Morphisms of Spaces results cited in the proof. Control the finite-fibre hypotheses and properness in the algebraic-space setting.

Source reference. Original reference 0AC7: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AC6: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 03I7: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BM: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02LN: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01WN: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03GN: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0262: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NX: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IJ: focused-seeds/geometry/S9-spaces-etale-splits-off-quasi-finite-spaces/ref-tags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $y \in |Y|$. Let $x_1, \ldots, x_n \in |X|$ mapping to y. Assume that*

1. *f is locally of finite type,*
2. *f is separated,*
3. *f is quasi-finite at $x_1, \ldots, x_n$, and*
4. *f is quasi-compact or Y is decent.*

Then there exists an étale morphism $(U, u) \to (Y, y)$ of pointed algebraic spaces and a decomposition

$$U \times_Y X = W \amalg V$$

into open and closed subspaces such that the morphism $V \to U$ is finite, every point of the fibre of $|V| \to |U|$ over u maps to an x_i, and the fibre of $|W| \to |U|$ over u contains no point mapping to an x_i.

Proof. Let $(U, u) \to (Y, y)$ be an étale morphism of algebraic spaces and consider the set of $w \in |U \times_Y X|$ mapping to $u \in |U|$ and one of the $x_i \in |X|$. By Decent Spaces, Lemma 0AC6 (if f is of finite type) or Decent Spaces, Lemma 0AC7 (if Y is decent) this set is finite. It follows that we may replace f by the base change $U \times_Y X \to U$ and $x_1, \ldots, x_n$ by the set of these w. In particular we may and do assume that Y is an affine scheme, whence X is a separated algebraic space.

Choose an affine scheme Z and an étale morphism $Z \to X$ such that $x_1, \ldots, x_n$ are in the image of $|Z| \to |X|$. The fibres of $|Z| \to |X|$ are finite, see Properties of Spaces, Lemma 03IJ (or the more general discussion in Decent Spaces, Section 03I7). Let $\{z_1, \ldots, z_m\} \subset |Z|$ be the preimage of $\{x_1, \ldots, x_n\}$. By More on Morphisms, Lemma 02LN there exists an étale morphism $(U, u) \to (Y, y)$ such that $U \times_Y Z = Z_1 \amalg Z_2$ with $Z_1 \to U$ finite and $(Z_1)_y = \{z_1, \ldots, z_m\}$. We may assume that U is affine and hence Z_1 is affine too.

Since f is separated, the image V of $Z_1 \to X$ is both open and closed (Morphisms of Spaces, Lemma 04NX). Set $W = X \setminus V$ to get a decomposition as in the lemma. To finish the proof we have to show that $V \to U$ is finite. As $Z_1 \to V$ is surjective and étale, V is the quotient of Z_1 by the étale equivalence relation $R = Z_1 \times_V Z_1$, see Spaces, Lemma 0262. Since f is separated, $V \to U$ is separated and R is closed in $Z_1 \times_U Z_1$. Since $Z_1 \to U$ is finite, the projections $s, t : R \to Z_1$ are finite. Thus V is an affine scheme by Groupoids, Proposition 03BM. By Morphisms, Lemma 03GN we conclude that $V \to U$ is proper and by Morphisms, Lemma 01WN we conclude that $V \to U$ is finite, thereby finishing the proof. □